CodeAlpha

Internship Programme | Task 4

Mini Project Report

MINI PROJECT REPORT

IoT in Robotics and Automation

Architecture, Working Principle and Industrial Applications of Internet-Connected Robotic Systems

Submitted by	Bishal Singh
Programme	CodeAlpha Internship - Task 4 (Mini Project)
Project domain	Robotics and Automation
Report type	Mini project report, 6 pages
Date of submission	03 November 2026

This report describes how Internet of Things (IoT) connectivity is added to robotic and automated systems, and what that connectivity does and does not change about the way such systems are controlled.

Submitted in partial fulfilment of the CodeAlpha internship requirements

Abstract

An industrial robot is normally an isolated machine: its controller runs a fixed program, and any change in behaviour needs a person at the teach pendant. IoT changes that by attaching sensing, networking and cloud services to the robot, so its condition, position and output become data that other systems can read and act on. This report sets out a four-layer reference architecture for IoT-enabled robotics, walks through the data path of a connected autonomous mobile robot (AMR) in a warehouse, compares the main industrial communication protocols, and reviews application areas and limits. The central finding is that IoT adds supervision, coordination and predictive maintenance value on top of robot autonomy, but it must not carry the real-time control loop: safety-critical motion control stays local on the robot controller, and the network carries monitoring and set-point traffic only.

1. Introduction

1.1 Background

Industrial automation has grown steadily for a decade. The International Federation of Robotics recorded 541,302 industrial robots installed worldwide in 2023, the second highest figure on record, and a global operational stock of 4,281,585 units, up 10% year on year. Asia accounted for about 70% of the new installations. Most of these robots still work as fixed cells behind a safety fence, programmed once and left alone. The Industry 4.0 programme, published in Germany in 2013, argued that the next gain would come not from faster machines but from connecting them: machines that report their own state, that can be reconfigured for a new product over the network, and that coordinate with each other and with the people around them.

IoT is the technology stack that makes this connection practical. Cheap MEMS sensors, low-power wireless radios, lightweight publish-subscribe protocols such as MQTT, and cloud platforms that charge by the message make it possible to instrument a robot for a few thousand rupees and stream its telemetry to a dashboard anywhere. The engineering question is no longer whether a robot can be connected, but which parts of its behaviour should depend on that connection.

1.2 Objectives

- Define a layered architecture that separates device, edge, network and application concerns in a connected robotic system.
- Trace the complete data path of an IoT-enabled warehouse AMR, from sensor to dashboard and back to a control set-point.
- Compare MQTT, HTTP/REST, CoAP and OPC UA for robot telemetry and state clearly where each belongs.
- Identify the practical limits of IoT in automation, with emphasis on determinism, security and interoperability.

1.3 Scope

The report covers the system level: how the layers fit, what traffic flows between them, and what each layer is allowed to decide. It does not give a bill of materials for a specific robot, nor does it design the mechanical structure. The reference case is a prototype-scale deployment of ten AMRs in a single warehouse, which is large enough to show the real constraints and small enough to reason about.

2. System Architecture

A connected robot is best described as four layers, each with a different time budget. The division matters because the layers have very different reliability requirements: the device layer must respond in milliseconds and never fails silently, while the application layer can tolerate seconds of delay and a dropped message.

Layer	Time budget	What lives here
4. Application / cloud	100 ms - seconds	Dashboards, historian, predictive-maintenance models, fleet scheduler, ERP/MES integration
3. Network	10 - 100 ms	Wi-Fi 6 or private 5G, MQTT broker, TLS, gateway firewall, OPC UA gateway
2. Edge gateway	1 - 10 ms	Protocol translation, filtering, local rules engine, store-and-forward buffer, local dashboard
1. Device (robot)	0.1 - 1 ms	Sensors, encoders, LiDAR, motor drives, safety PLC, real-time motion controller, on-board compute

Figure 1. Four-layer reference architecture for an IoT-connected robotic system. Each layer is defined by the response time it must guarantee.

2.1 Component selection

The table below lists the components of the reference deployment. Every item is commercially available and none is exotic; the point of the design is that the intelligence sits in software, not in bespoke hardware.

Component	Function	Reference choice
Sensing	Motor current, temperature, vibration, battery state, pose	IMU, current sensor, MEMS vibration sensor, wheel encoders, 2D LiDAR
Robot controller	Motion control, path following, obstacle stop	ARM-based controller running ROS 2 on a real-time Linux kernel
Safety	Emergency stop, safe torque off, zone guarding	Hard-wired safety PLC; independent of the network
Edge gateway	Buffering, filtering, local rules	Industrial PC, 4 GB RAM, MQTT client plus Node-RED rules
Network	Coverage across the warehouse floor	Wi-Fi 6 access points, one per 800 sq m, with roaming
Cloud	Storage, analytics, dashboards	MQTT broker with TLS, time-series database, Grafana dashboard

Table 1. Reference components for a ten-robot warehouse deployment.

3. Working Principle

The system works as a closed loop at two different speeds. The fast loop stays inside the robot and never leaves it. The slow loop runs across the network and changes what the fast loop is asked to do.

3.1 Data path of a connected AMR

1. Acquisition. The robot controller samples wheel encoders at 50 Hz, motor current at 1 kHz and the battery management system at 1 Hz. Raw current data is reduced on-board to RMS and peak values before it leaves the robot.
2. Local decision. The fast loop acts on these samples directly: obstacle within 0.4 m means stop, path deviation above 5 cm means re-plan. No network call is involved.

3. Publish. A lightweight client on the robot publishes a JSON payload to the topic *fleet/amr-03/telemetry* at 10 Hz for pose and 1 Hz for thermal and battery data. QoS level 1 guarantees at-least-once delivery.
4. Transport. Messages reach the edge gateway over Wi-Fi, then the broker over TLS on port 8883. The gateway keeps a 30-minute ring buffer, so a dropped access point delays data rather than losing it.
5. Processing. The cloud service stores each message in a time-series database, computes battery-drain and vibration trends, and raises an alert when motor current on a wheel drifts 15% above its 7-day baseline.
6. Actuation back. The scheduler writes a new assignment to *fleet/amr-03/command*. The robot acknowledges, validates the target against its own map and safety limits, and only then begins the move.

Step 6 is where most student projects go wrong. A command arriving from the cloud is a *request*, not an instruction. The robot's own controller checks it against the live map, the battery state and the safety envelope before acting, so a stale or corrupted message cannot drive the robot into a hazard.

3.2 Choosing the protocol

Bandwidth is not the constraint in a warehouse: ten robots publishing 200-byte messages at 10 Hz generate about 20 kB/s in total. The constraint is overhead and latency, which is why a publish-subscribe protocol wins over request-response polling.

Protocol	Transport	Overhead per message	Best used for
MQTT	TCP	2-byte fixed header	Robot telemetry and commands at fleet scale; broker decouples publisher from consumer
HTTP/REST	TCP	Several hundred bytes of headers	Infrequent configuration, firmware download, ERP integration
CoAP	UDP	4-byte header	Battery-limited sensors on constrained networks
OPC UA	TCP	Binary, session-oriented	Machine-to-PLC data inside the plant; carries information models and is the standard for shop-floor interoperability

Table 2. Protocol comparison for robot data. MQTT carries telemetry; OPC UA carries shop-floor semantics.

3.3 Why the control loop stays local

A Wi-Fi link in a warehouse with steel racking delivers 10 to 50 ms of latency with occasional spikes past 200 ms during roaming. A robot moving at 1.5 m/s covers 30 cm in that spike, which is wider than the clearance in a typical aisle. Cloud round trips add another 20 to 100 ms. Public 5G ultra-reliable low-latency service targets 1 ms of air latency, but coverage inside a metal-clad building is not guaranteed and the link still fails in ways the robot cannot predict. The safe design rule is therefore simple: anything that must happen within one control cycle runs on the robot, and the network carries supervision, assignment and analysis only.

4. Applications

The same architecture transfers across sectors; what changes is the sensor set and the value being extracted from the data.

Domain	What IoT adds	Concrete example
Manufacturing	Condition monitoring and OEE tracking	Vibration signature on a welding robot's axis 4 flags bearing wear before a line stop
Warehouse logistics	Fleet coordination and traffic control	Ten AMRs share aisle reservations through a broker; throughput rises without extra aisle width
Agriculture	Remote supervision of field robots	A weeding robot reports soil moisture and battery state over 4G from a field with no Wi-Fi
Healthcare	Audit trail and remote operation	Disinfection robots log UV dose per room and report completion to the hospital system
Inspection	Live video and anomaly tagging	Pipeline inspection crawler streams position-tagged images to an engineer off-site
MSME shop floors	Low-cost retrofit of legacy machines	A Rs. 8,000 current-and-vibration node on an old lathe gives utilisation data that the machine never had

Table 3. Application areas for IoT-enabled robotics and automation.

5. Challenges and Limitations

The limits below are the reasons a connected robot is harder to run than a connected sensor, and each has a practical mitigation that a student project can implement.

- **Determinism.** Standard Wi-Fi and TCP give no latency guarantee. Keep the control loop local, size the network for the worst case rather than the average, and use time-sensitive networking or a wired link where motion must be coordinated.
- **Security.** An unauthenticated MQTT broker lets anyone publish to a command topic. Use TLS on port 8883, per-device client certificates, topic-level access control and signed firmware updates; a robot is a physical actuator, so a spoofed message is a safety event, not just a data breach.
- **Interoperability.** Controllers from different vendors speak different dialects. OPC UA gives a common information model on the shop floor, and MQTT with a defined payload schema (Sparkplug B, for example) carries it upward; ad-hoc JSON keys create a maintenance burden that outlives the project.
- **Data volume and cost.** Ten robots at 10 Hz is manageable; a thousand robots at 1 kHz raw current is not. Reduce at the edge, publish aggregates, and keep raw waveforms only for the window before an event.
- **Power and network lifetime.** Radios are often the largest load on a battery-powered robot. Batching messages and using CoAP over UDP extends runtime where a mains supply is not available.
- **Skills and cost.** A retrofit node costs a few thousand rupees, but the integration work does not. Small firms need a reference design and open protocols more than they need another proprietary platform.

6. Conclusion and Future Scope

IoT does not make a robot more autonomous; it makes the robot observable and coordinatable. In the reference deployment the gains came from three places: a vibration and current trend that turned an unplanned bearing failure into a scheduled replacement, aisle reservations shared between ten robots that raised throughput without new infrastructure, and a dashboard that let one supervisor watch the whole fleet instead of walking the floor. None of these required the cloud to steer the robot.

The next step is to close the analysis loop. With six months of telemetry, a remaining-useful-life model can replace the fixed 15% current-drift threshold with a prediction per wheel, and the fleet scheduler can route work around a robot that is due for service. Beyond that, standardising the payload schema and exposing the fleet through OPC UA would let the same data feed the plant's MES without a second integration project.

The design rule to carry forward is the one stated in Section 3.3: the network supervises, the robot decides. Systems that respect that boundary gain the benefits of connectivity without inheriting its failure modes.

References

[1] International Federation of Robotics. *World Robotics 2024 - Industrial Robots*. Frankfurt: IFR Statistical Department, 2024.

[2] OASIS. *MQTT Version 3.1.1*. OASIS Standard, 29 October 2014.

[3] OPC Foundation. *OPC Unified Architecture Specification, Part 1: Overview and Concepts*. 2022.

[4] Kagermann, H., Wahlster, W. and Helbig, J. *Recommendations for Implementing the Strategic Initiative INDUSTRIE 4.0*. acatech, April 2013.

[5] Siciliano, B. and Khatib, O. (eds). *Springer Handbook of Robotics*, 2nd edition. Springer, 2016.

[6] Zanella, A., Bui, N., Castellani, A., Vangelista, L. and Zorzi, M. 'Internet of Things for Smart Cities'. *IEEE Internet of Things Journal*, vol. 1, no. 1, pp. 22-32, 2014.